

# RootLens: Visual Analytics for Multi-Source Industrial Anomaly Detection and Traceable Root-Cause Analysis

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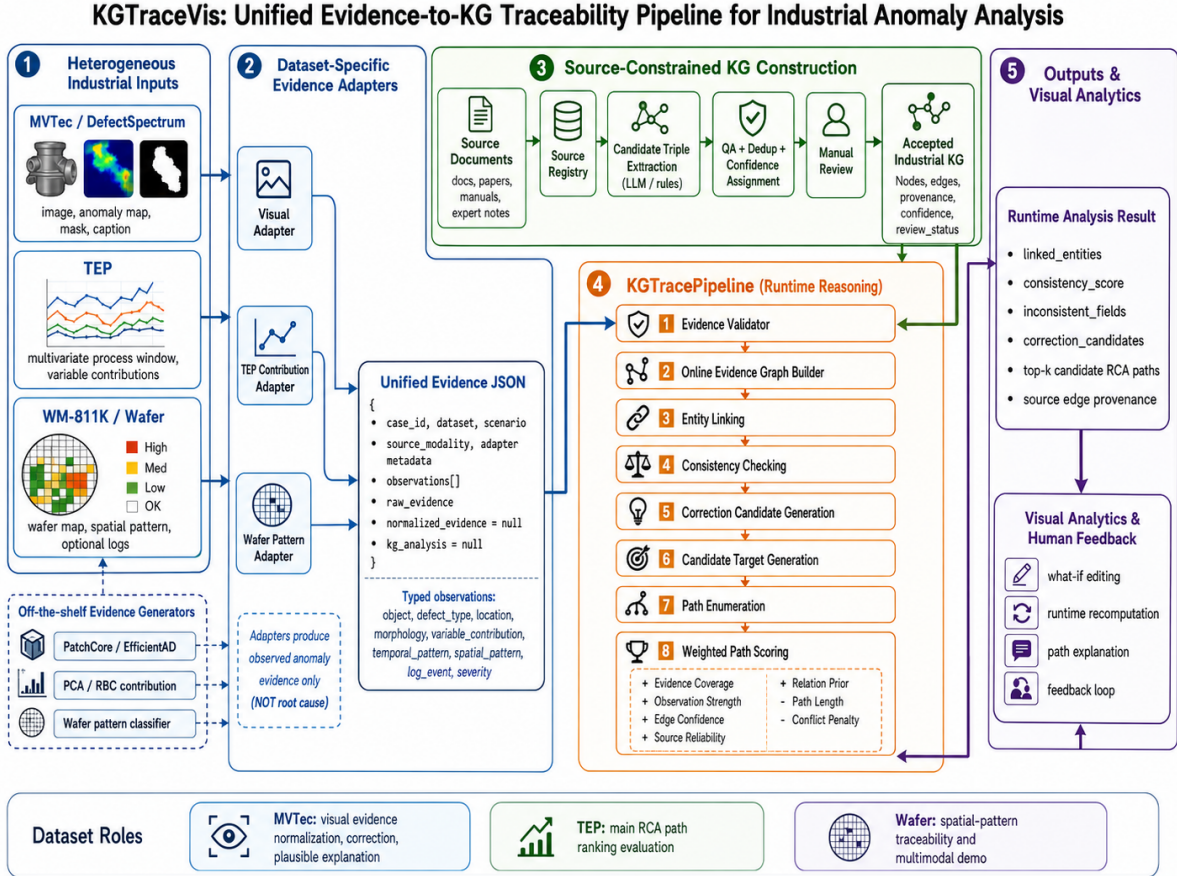


Figure 1: RootLens supports traceable industrial anomaly analysis by organizing multi-source evidence, candidate root-cause paths, and source-grounded explanations in an integrated visual workflow.

## ABSTRACT

Industrial anomaly management requires moving beyond detecting abnormal signals toward traceable root-cause analysis that combines observations, process knowledge, logs, documents, and expert judgment. Existing detection models, root-cause analysis algorithms, and industrial knowledge graph construction methods are often tied to single modalities or scenarios, making it difficult to preserve source traces and support human verification. We present

RootLens, an evidence-centered visual analytics framework that maps outputs from images, time series, logs, and documents into a unified representation of evidence, candidate knowledge graph entities, confidence scores, and source traces. RootLens combines source-grounded knowledge graphs, pluggable root-cause reasoning, candidate path visualization, and human-in-the-loop graph management to support an analysis workflow from anomaly observations to root-cause hypotheses. Case studies on process faults and visual defect scenarios, together with expert feedback, demonstrate the framework’s potential for organizing cross-source clues and examining plausible, explainable root causes.

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## 1 INTRODUCTION

## 2 RELATED WORK

### 2.1 Industrial Anomaly Detection and Root Cause Analysis

### 2.2 Knowledge Graphs and LLMs for Industrial Knowledge

### 2.3 Visual Analytics for Diagnosis, Provenance, and Human-AI Decision Making

## 3 DOMAIN PROBLEM AND DESIGN REQUIREMENTS

### 3.1 Industrial RCA Workflow

Following established design-study practice, a visual analytics system for industrial root-cause analysis (RCA) should first be grounded in the domain workflow rather than in a particular interface or algorithmic component [12, 16]. In industrial anomaly analysis, detecting an abnormal sample is rarely the end of the task. Analysts often receive heterogeneous outputs from visual inspection models, process-variable monitoring, wafer spatial-pattern analysis, log events, and related operational records. These outputs differ in scale, semantics, uncertainty, and temporal or spatial context. Prior studies on industrial data fusion and anomaly exploration have similarly shown that diagnosis requires connecting heterogeneous observations with domain knowledge and human judgment, rather than treating anomaly scores as self-contained explanations [2, 19, 22].

A practical RCA workflow therefore proceeds through a sequence of interpretation and verification steps. First, analysts collect anomaly observations from one or more sources and interpret them as concrete evidence, such as a defect region, an abnormal variable segment, a wafer-level spatial pattern, or an event record. Second, these observations must be grounded to domain entities, including objects, process units, variables, locations, morphology terms, or event concepts. Third, analysts inspect whether the linked observations are consistent with process knowledge and whether alternative interpretations or corrections are needed. Fourth, they explore candidate root-cause paths or ranked hypotheses, together with source traces and supporting evidence. Finally, they record feedback by confirming, rejecting, or marking candidates for further review. Similar post-detection workflows have been observed in visual analytics systems for anomaly diagnosis and knowledge-graph-supported RCA, where analysts repeatedly move between machine-generated candidates, contextual evidence, and human verification [22, 11, 21].

The main difficulty is that these steps are often fragmented across modality-specific tools, scenario-specific models, and disconnected inspection records. Such fragmentation makes it hard to compare observations across sources, preserve the provenance of each clue, understand the confidence and uncertainty of candidate links, and maintain a reproducible record of human decisions. In addition, a candidate path in an RCA graph is not a verified cause by itself: analysts still need to inspect why the path was suggested, which evidence supports it, and which assumptions or graph relations remain uncertain. These challenges motivate RootLens as a post-detection visual analytics system for organizing heterogeneous anomaly observations into source-grounded, provenance-preserving, and graph-linked evidence that supports reviewable candidate RCA.

### 3.2 Design Requirements

The workflow above motivates five design requirements for RootLens.

**R1. Unified multi-source evidence representation.** RootLens must represent heterogeneous observations from images, time series, logs, wafer maps, and related records in a common post-detection evidence form so that they can enter the same downstream

RCA workflow. This requirement allows analysts to compare observations that originate from different detectors or operational sources without forcing all data into a single modality-specific format. It also provides the basis for connecting anomaly observations to shared domain entities and process knowledge [2, 19].

**R2. Preservation of provenance, confidence, and raw context.** RootLens must preserve each observation’s source trace, confidence, modality-specific context, and review boundary rather than collapsing heterogeneous evidence into opaque labels. Provenance and uncertainty are central to trustworthy visual analytics because analysts need to know where a claim comes from, how reliable it is, and whether it should be treated as an observation, a candidate relation, or a confirmed decision [13, 14]. For RCA, such preservation is especially important because evidence from different sources may be incomplete, noisy, or mutually inconsistent.

**R3. Pluggable and scenario-aware RCA reasoning.** RootLens must support multiple RCA strategies without assuming one universal detector, one fixed causal model, or one scenario-independent reasoning procedure. Different industrial settings may require graph-path search, process-specific RCA models, similarity-based ranking, or other reasoning providers. The system should therefore align their outputs into a common representation of candidate hypotheses while keeping the underlying reasoning strategy explicit and reviewable. This requirement follows prior work showing that causal and root-cause analysis is often exploratory and requires human interpretation rather than fully automatic verification [11, 21].

**R4. Linked analysis across evidence, graph, and candidate paths.** RootLens must let analysts move between evidence items, linked knowledge-graph entities, consistency or correction results, candidate explanation paths, and ranked root-cause hypotheses within one connected workflow. A ranked list alone is insufficient for RCA because analysts need to inspect how a hypothesis is grounded in evidence, how it traverses domain knowledge, and which neighboring entities or alternative paths may explain the same anomaly. Linked visual analysis can reduce the gap between post-detection observations and graph-level reasoning by making the evidence–entity–path relationship explicit and interactive [22, 11].

**R5. Human-in-the-loop review and reproducible curation.** RootLens must support lightweight human confirmation, rejection, and curation of candidate outputs and graph materials while preserving reproducibility and claim boundaries. Candidate graph relations and candidate root-cause hypotheses should not automatically become trusted facts; instead, analysts should be able to inspect their supporting evidence, record feedback, and distinguish reviewed decisions from unresolved candidates. This requirement reflects the role of visual analytics in knowledge generation, where human expertise, system-generated suggestions, uncertainty awareness, and feedback jointly shape the analysis process [15, 14, 11].

Together, these requirements motivate the evidence-centered RCA framework in Section 4 and the RootLens visual analytics system in Section 5. Table 1 summarizes how the requirements are translated into RootLens design responses and where they are examined later in the paper.

## 4 EVIDENCE-CENTERED RCA FRAMEWORK

RootLens frames root-cause analysis (RCA) as an evidence-centered visual analytics workflow rather than as a new anomaly detector or a universal causal model. Industrial analysis often combines anomaly observations and contextual knowledge from multiple sources, including image-based inspection outputs, process-variable signals, wafer-map patterns, logs, operator records, and domain documents. These sources differ in modality, confidence semantics, spatial or temporal reference, and provenance. The role of the framework is to turn heterogeneous anomaly outputs into comparable and traceable evidence, connect them with source-grounded

Table 1: Design requirements derived from the industrial RCA workflow.

| Requirement                               | Problem motivation                                                                       | RootLens design response                                                                            | Evidence in paper                                                             |
|-------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| R1 Unified multi-source evidence          | Observations come from incompatible model- or modality-specific formats.                 | Evidence-centered normalization after detection.                                                    | Section 4; cases across TEP, MVTec, and wafer scenarios in Section 6.         |
| R2 Provenance and confidence preservation | Analysts need to know where each clue came from and how reliable it is.                  | Source-grounded evidence fields, reviewable graph relations, confidence, and claim boundaries.      | Sections 4–5; path provenance in Section 6.                                   |
| R3 Pluggable RCA reasoning                | Different scenarios require different reasoning strategies.                              | Generic graph-path reasoning and scenario-aware reasoning providers with aligned candidate outputs. | Reasoning layer in Section 4; TEP case in Section 6.                          |
| R4 Linked graph–evidence–path analysis    | Analysts must traverse between observations, domain context, and candidate explanations. | Integrated evidence, graph, and path analysis with candidate ranking and interactive inspection.    | System workflow in Section 5; usage scenarios in Section 6.                   |
| R5 Human-in-the-loop curation             | Candidate explanations and candidate graph relations require human judgment.             | Feedback, review targets, graph workshop, and source-grounded construction management.              | Review workflow in Section 5; expert feedback and discussion in Sections 6–7. |

KG knowledge, and produce reviewable root-cause hypotheses for the visual analytics system in Section 5.

#### 4.1 Framework Overview and Design Rationale

Guided by the design requirements in Section 3, RootLens separates detection outputs, evidence normalization, source-grounded knowledge, runtime reasoning, and visual review. The rationale is to keep modality-specific detectors and data processors replaceable while making their outputs comparable, traceable, and reviewable in a common RCA workflow. This separation also avoids treating a detector output, dataset label, or extracted text fragment as a causal explanation before it has been linked to evidence and checked against source-grounded knowledge.

The framework follows the pipeline:

*producer* → *adapter* → *runtime reasoning pipeline* → *reviewable outputs*.

Model-aware producers transform raw inputs or model outputs into normalized records. For instance, a producer may summarize an image detector’s heatmap and mask, a process model’s variable contributions, or a wafer classifier’s spatial pattern prediction. Model-independent adapters then map these records into the unified Evidence contract. Adapters may be dataset-aware in what they read, but they are not responsible for RCA reasoning. Their role is to preserve what was observed, how it was produced, and where it can be traced back.

The unified evidence object enters the RootLens runtime reasoning pipeline, implemented as `KGTracePipeline`. The pipeline performs evidence-to-KG analysis and a final scenario-aware RCA stage. It returns an `AnalysisResult` that exposes linked entities, consistency information, correction candidates, candidate root-cause paths, ranked root-cause hypotheses, source-edge provenance, and feedback state. These outputs form the framework-level contract consumed by the visual analytics system, while the specific visual encodings and interactions are described in Section 5.

#### 4.2 Evidence Abstraction

The central data abstraction in RootLens is `Evidence`. It represents an analysis case as source-grounded observations rather than as a detector-specific prediction. At the top level, an evidence object records case identity and modality context, observed anomaly descriptors, raw provenance, canonical observations, runtime analysis state, and human feedback state. This structure allows the framework to normalize evidence for reasoning while keeping dataset- and model-specific material available for inspection.

The case envelope identifies the dataset, source, timestamp, and modality context. Observed anomaly descriptors summarize what was detected or reported, such as object, anomaly type, location,

morphology, severity, and confidence. Raw provenance retains source-specific material: visual cases may keep heatmap paths, masks, bounding boxes, or geometry; process cases may keep variables, contribution values, and time windows; wafer cases may keep spatial-pattern descriptors and classifier metadata. In this way, RootLens does not force all modalities into one raw representation. Instead, it defines a common reasoning surface over heterogeneous observations.

The canonical reasoning unit is `EvidenceObservation`. Each observation has a stable identifier, an evidence facet, a value or name, confidence, a source reference, a raw-field reference, an optional time window, and metadata. This abstraction makes heterogeneous evidence items referenceable in the same reasoning layer. A visual anomaly region, a high-contribution process variable, a wafer spatial pattern, and a log event can all be represented as observations, linked to KG entities, checked for consistency, and exposed as review targets.

Adapters instantiate this abstraction conservatively. Visual adapters normalize image-level anomaly scores, localization geometry, morphology, location, severity, and confidence; these records are treated as visual anomaly or localization evidence, not as semantic root-cause labels. The TEP adapter represents process evidence through variables, variable contributions, process metadata, time windows, severity, and confidence. The wafer adapter represents spatial-pattern evidence, zones, morphology, severity, wafer-map descriptors, and classifier provenance. Across scenarios, adapters produce observed anomaly evidence only. They do not emit root causes, ranked paths, or trusted KG conclusions.

#### 4.3 Source-Grounded Knowledge Supply

RootLens pairs the evidence abstraction with a source-grounded knowledge supply layer. Rather than assuming a complete industrial KG, the framework uses task-oriented KG snapshots that can be constructed, checked, reviewed, and loaded for runtime reasoning. Domain documents, process tables, curated records, and structured inputs enter this layer as registered sources. Extractor modules generate draft entities and relations, which are then cleaned, validated, and prepared as candidate KG artifacts.

This construction process distinguishes candidate knowledge from reviewed knowledge. Draft entities and relations retain source identifiers, extractor information, scenarios, supporting evidence spans or text, confidence, status, and metadata. Their status indicates whether an item is still a draft, has been accepted, rejected, or published. This distinction is especially important for LLM- or document-extracted triples: in RootLens, such outputs are reviewable candidates unless they have been source-attached and reviewed. They are not treated as trusted industrial facts by default.



The runtime KG preserves provenance at the edge level. Nodes carry identifiers, names, labels, aliases, scenarios, and descriptions. Edges carry the head, relation, tail, scenario, source, supporting evidence, confidence, weight, review status, and feedback counters. As a result, a candidate root-cause path is not merely a graph traversal result. It is a source-grounded explanation structure whose supporting edges can be inspected, questioned, accepted, or rejected. For a selected case, the runtime pipeline retrieves a scenario-scoped graph snapshot that includes the relevant scenario layer and shared knowledge layer.

#### 4.4 Evidence-to-KG Reasoning Workflow

All cases enter the reasoning pipeline through the unified Evidence contract and pass through a shared set of evidence-to-KG stages: scenario-scoped KG retrieval, entity linking, consistency checking, and correction candidate generation. These stages use the same runtime contracts, but their applicable observations, relation checks, and candidate corrections depend on the scenario-specific evidence and KG snapshot.

First, the pipeline retrieves the KG snapshot relevant to the evidence case. This step scopes reasoning to the case scenario while retaining shared cross-scenario knowledge when available. Second, entity linking maps canonical observations to candidate KG nodes. Linkable facets include objects, anomaly types, locations, morphologies, process variables, and log events. The linker uses identifiers, names, aliases, partial matches, and fuzzy matches, and it records ambiguous matches instead of forcing low-confidence links.

Third, the consistency checker compares linked evidence fields against KG relation rules. These checks are relation-based rather than interface-specific. For example, the KG may encode whether an anomaly type is associated with a morphology or location, whether a process variable belongs to a unit or measurement location, or whether an anomaly type is associated with a log event. The resulting consistency score summarizes entity-link coverage and relation-rule agreement, while inconsistent fields are retained as reviewable evidence-to-KG conflicts.

Fourth, when relation checks reveal conflicts, the pipeline generates correction candidates from KG neighborhoods. A correction candidate records the original field or value, a suggested entity or value, a score, a reason, and supporting edge provenance. The pipeline does not mutate the original evidence. Corrections are presented as reviewable alternatives that can inform analyst decisions, evidence revision, or KG curation.

#### 4.5 Scenario-Aware RCA and Reviewable Outputs

The final stage of the reasoning pipeline performs scenario-aware RCA reasoning. The shared evidence-to-KG stages provide linked entities, consistency checks, and correction candidates as review context. The final RCA stage then selects an appropriate reasoning strategy while preserving a unified output contract. In graph-oriented scenarios, the generic reasoner uses linked entities and the runtime KG snapshot. In TEP, the Root-KGD reasoner consumes Evidence-derived variable contributions and dynamic features together with prebuilt Root-KGD graph and scoring assets.

For default visual, wafer, or other graph-oriented cases, the generic reasoner ranks candidate paths from linked evidence entities to root-cause-like targets, such as root-cause, cause-category, or fault-type nodes. Its path score combines relation confidence, evidence match, and a length penalty:

$$\text{Score}(P) = \alpha \cdot \text{Conf}(P) + \beta \cdot \text{EvidenceMatch}(P) - \gamma \cdot \text{Length}(P).$$

This score is a ranking heuristic for prioritizing source-grounded candidate paths. It should not be interpreted as proof of causal correctness.

For TEP, RootLens uses a Root-KGD reasoner as the primary quantitative RCA scenario. The reasoner uses Evidence-derived variable contributions and dynamic features, together with the pre-built Root-KGD graph and scoring assets. It does not rely on `linked_entities` for RCA scoring; linked entities and consistency checks remain review context. TEP fault numbers may be present as evaluation references or metadata, but they are not scoring inputs. This design allows TEP to support RCA evaluation while preserving the same framework boundary between observed evidence, runtime reasoning, and visual review.

Different RCA strategies return the same output contract. The `AnalysisResult` exposes `top_k_paths`, `ranked_root_causes`, supporting edges and evidence, scoring method, scoring details, review status, and feedback state, together with the linked entities, consistency score, inconsistent fields, and correction candidates computed earlier. Each candidate root-cause path retains its node and relation sequence, score components, supporting evidence, and source-edge records. Each ranked root-cause hypothesis retains its candidate identity, rank, confidence, evidence match, explanation paths, supporting edges, and review status.

The scenario boundaries are intentionally conservative. TEP is the main RCA reasoning and evaluation scenario. MVTec-style cases support visual evidence normalization and plausible candidate explanations, but they should not be interpreted as verified factory RCA. WM811K/wafer cases support spatial-pattern evidence and candidate traceability; process-level RCA claims require reviewed process references. Across all scenarios, RootLens presents RCA outputs as reviewable hypotheses rather than as discovered ground truth. Feedback can target paths, edges, entity links, correction candidates, or KG draft items, but in the framework it remains review state rather than an automatic mutation of production KG artifacts.

### 5 VISUAL ANALYTICS SYSTEM

RootLens instantiates the evidence-centered RCA framework as a linked visual analytics workbench. Rather than exposing the reasoning pipeline as a black-box result table, the system organizes uploaded or replayed runs, evidence observations, graph entities, candidate explanation paths, provenance records, and human review state into coordinated workspaces. The design follows the classical overview-filter-detail pattern and the broader interaction principles of visual analysis systems [17, 7], while adapting them to the industrial RCA setting where analysts must compare heterogeneous evidence, inspect plausible hypotheses, and keep the boundary between system suggestions and human verification explicit. In the current implementation, the workflow is organized around three workspaces: evidence review, graph/path exploration, and provenance-centered feedback and graph-material curation.

#### 5.1 Evidence Overview and Detection Details

The evidence workspace serves as the entry point from raw anomaly outputs to reviewable case context. Analysts can start from uploaded runs or replayed runtime assets, inspect run history and workflow progress, and select a case from a case table for deeper review. This case-centered entry avoids forcing analysts to reason directly over disconnected detector outputs. Instead, observations from different sources are normalized into comparable evidence items that can be browsed by source, modality, time, confidence, and case association.

The main interaction in this workspace is progressive narrowing. A run-level overview gives analysts a compact summary of available cases and observations; graph, facet, and keyword filters then help isolate evidence related to a component, variable, document cue, or candidate symptom. When available, modality-specific details such as image regions, masks, heatmap-like cues, time-series windows, variable names, and textual snippets are displayed close

to the corresponding evidence item. This design is consistent with prior visual analytics systems for anomaly inspection, which emphasize that automated detection results must be made inspectable and correctable by domain experts rather than treated as final labels [23, 9]. RootLens differs from systems that focus mainly on evaluating anomaly detectors or reviewing multivariate time-series structures [24, 5]: here, evidence inspection is used to establish the case context that will later constrain graph-based hypothesis review.

The selected case and evidence filters are preserved as shared analysis state. As a result, analysts can move from the evidence workspace to the graph workspace without losing the current run, case, or observation context. This continuity helps realize the requirements for multi-source evidence integration and case-centered inspection by turning heterogeneous detector and extraction outputs into stable analytical objects before any root-cause hypothesis is examined.

## 5.2 Knowledge Graph Reasoning View

The graph exploration workspace is the main view for inspecting how evidence is connected to entities, relations, and candidate root-cause hypotheses. It coordinates a global knowledge graph context with a case-specific explanation subgraph derived from the current `path.graph`. The global graph provides background structure about assets, symptoms, variables, documents, and operational concepts, while the case-specific subgraph focuses attention on the entities and relations that participate in the current explanation path. This separation follows prior work on multivariate and heterogeneous graph exploration, where analysts benefit from moving between broad graph context and focused aggregates or trails [20, 3].

RootLens presents candidate root-cause hypotheses as ranked, reviewable outputs rather than verified causal labels. Each candidate card exposes its rank, score, scoring method, and role in the current case, and is linked to the corresponding nodes and edges in the path graph. Selecting a candidate highlights the plausible explanation path and lets analysts traverse from symptoms to intermediate evidence and candidate causes. This linked inspection design is related to recent work on guided knowledge graph exploration [25] and KG-based visual analytics for cloud anomaly investigation [10], but RootLens emphasizes source-grounded evidence, explicit claim boundaries, and reviewable industrial RCA outputs across multiple evidence modalities.

The view also supports run and case switching inside the graph workspace, allowing analysts to compare whether similar entities or paths recur across cases. Importantly, the graph view does not claim to prove causality. Its role is to make the reasoning output visually inspectable: analysts can see which entities were connected, which path was suggested, which scoring signal was used, and where the system’s claim boundary begins and ends. This design makes the knowledge graph an interface for hypothesis inspection and sense-making, not an automatic replacement for domain judgment.

## 5.3 Provenance and Human Feedback

The third part of the system keeps provenance and feedback explicit. For candidate paths, edges, and entity links, RootLens surfaces source-grounded provenance so analysts can inspect why a relation or path was suggested before recording feedback. Provenance is central to visual analytics because it supports recall, communication, verification, and accountability throughout the analysis process [13, 6]. In RootLens, provenance records connect graph elements back to evidence snippets, source materials, document locations, or model outputs when such metadata is available. This helps analysts distinguish a relation supported by concrete evidence from one that should remain a tentative graph association.

Human feedback is represented as append-only review state over bounded review targets. The strongest supported targets are paths, edges, entity links, and correction records; feedback on candidate

root-cause cards is therefore described conservatively as part of the review workflow rather than as a fully automatic write-back mechanism. Analysts can confirm, reject, annotate, or correct selected targets, and the resulting review state remains associated with the current run and case. This design follows the broader visual analytics view that sensemaking systems should help analysts externalize findings, hypotheses, and reasoning traces [18], while also recognizing that user feedback can guide analytical systems only when the semantics of the interaction are explicit [4].

RootLens further extends provenance-centered review with a graph-material workspace. Source-grounded materials can be searched, filtered, sorted, uploaded, and selected for graph construction; build lists, build details, and recipe-style forms support lightweight orchestration of graph construction tasks. This workspace is not a full production KG editor and does not automatically mutate the runtime KG based on feedback. Instead, it provides review-oriented graph management and draft construction support, aligning with recent observations that practical knowledge graph systems require better support for builders, analysts, provenance, and collaborative curation [8]. The use of knowledge graphs as curation resources also relates to systems such as CAVA, which integrate KG-backed augmentation with visual analysis workflows [1].

Together, these views turn the evidence-centered RCA framework into a connected review workflow that supports hypothesis inspection, provenance tracing, and lightweight human-in-the-loop graph curation. RootLens supports both reproducible replay over generated runtime assets and live backend integration, but the analyst-facing goal remains the same: to keep multi-source evidence, graph reasoning outputs, and human verification tightly linked while avoiding overclaiming candidate hypotheses as verified root causes.

## 6 CASE STUDIES AND EVALUATION

We evaluate RootLens through three complementary case studies and a planned formative expert-feedback protocol. The cases are intentionally not treated as equally strong ground-truth benchmarks: they cover different evidence strengths and test different segments of the evidence-centered workflow introduced in Section 4. The Tennessee Eastman Process (TEP) provides the primary quantitative evaluation because process-fault references allow RCA ranking to be scored. The WM811K wafer case evaluates spatial-pattern evidence normalization and source-grounded traceability; it does not provide verified process RCA in our setting. MVTec-style data evaluates visual evidence normalization, transparent object selection, KG completion, and plausible explanation review; it does not provide factory RCA labels. The planned expert-feedback protocol assesses whether unified evidence, source traces, candidate paths, and KG review targets support analyst sensemaking beyond isolated detector outputs.

Across all cases, RootLens outputs are described as ranked candidate hypotheses and source-grounded paths, not as causal proof. Quantitative metrics are reported only where suitable evaluation references exist; elsewhere we report traceability and reviewability evidence.

**KG artifact check.** Before the case studies, we ran a quality-assurance check over the paper-facing KG artifact set used across all three evaluation scenarios. The artifact set contains 211 nodes and 602 edges, with zero blocking issues and 22 warnings (18 isolated-node warnings and four reviewed low-confidence warnings). We report these warnings rather than suppressing them: they identify curation opportunities and source-quality issues that remain visible in the review workflow.

### 6.1 Case 1: Tennessee Eastman Process RCA

**Evaluation task.** The Tennessee Eastman Process (TEP) [?] is a widely used chemical-process benchmark offering 21 fault types

Table 2: TEP RCA results. Fault references are used only for post-hoc metric computation, not as scoring inputs. The *Abl.* column reports top-1 stability after removing `fault_number` and `fault_id` from producer records. Fault 21 was requested but absent from the local raw file; the broad fault-set run is the primary result and the sanity-check row is supplementary.

| Setting         | Faults             | Cases | Top-1 | Top-3 | Top-5 | MRR    | Path | Abl. |
|-----------------|--------------------|-------|-------|-------|-------|--------|------|------|
| Broad fault-set | 1–20 (req. 1–21)   | 20    | 0.80  | 0.90  | 0.90  | 0.8417 | 0.90 | 1.00 |
| Sanity check    | 1–8 ( $\times 2$ ) | 16    | 1.00  | 1.00  | 1.00  | 1.00   | 1.00 | 1.00 |

with distinct variable signatures. We use TEP as the primary quantitative RCA case because its process-fault references can be used after reasoning to evaluate ranked root-cause candidates. Raw TEP records are processed by a TEP producer that extracts window-level variable contributions and dynamic process features. The TEP adapter converts these records into the unified Evidence contract, which enters the KGTracePipeline. The TEP Root-KGD reasoner consumes the Evidence-derived contributions together with prebuilt Root-KGD graph and scoring assets, returning ranked root-cause candidates and top- $k$  support paths through the same output contract used in all other scenarios.

**Fault identifiers as post-hoc references.** Fault identifiers may be present in producer records as metadata, but they are treated as post-hoc evaluation references rather than RCA scoring inputs. To audit this boundary, we reran the full workflow after removing explicit `fault_number` and `fault_id` fields from producer records. The top-1 prediction remained stable in all 20 audited cases. This reduces concern that explicit benchmark labels leak into runtime rankings, but does not establish causal correctness.

**Metrics.** We report *top-k RCA accuracy* (fraction of cases in which the expected root-cause candidate appears in the top- $k$  ranked hypotheses), *mean reciprocal rank* (MRR), *path hit rate* (whether any top- $k$  path includes or targets the expected candidate), and *label-ablation stability* (top-1 stability after explicit fault identifiers are removed from records).

**Broad fault-set evaluation.** The main TEP run requested faults 1–21, with one case per observed fault and `top_k=5` [VERIFY: final configuration]. The local raw file contained faults 1–20; fault 21 was not observed. The evaluation therefore covers 20 cases. As shown in Table 2, the TEP RCA workflow returned 0.80 top-1 accuracy, 0.90 top-3 and top-5 accuracy, 0.8417 MRR, 0.90 path hit rate, and 1.00 label-ablation stability across all 20 audited cases. We select this run as the primary result because it covers the full observed fault range and exposes both successful rankings and current limitations.

A supplementary sanity check on faults 1–8 with two cases per fault (16 cases) returned 1.00 on all metrics. We treat this as a controlled sanity check rather than the headline evaluation because the broad fault-set run more clearly exposes failure modes and asset-coverage boundaries.

**Failure and near-miss analysis.** The broad fault-set run exposes four cases that do not achieve a top-1 hit (Table 3). Faults 10 and 15 are near misses: the expected candidates appear within the top-3 list and their paths are retrieved, but a recurrent feed-temperature candidate ranks first under current scoring. Fault 18 is a miss under the current Root-KGD assets and dynamic features. Fault 20 is an asset-coverage gap: the workflow falls back to an unresolved anchor because no Root-KGD fault node is available in the current KG snapshot. In the review workflow, an analyst can compare the ranked candidates, inspect the source-grounded support paths, and identify whether an error reflects scoring ambiguity, feature limitations, or missing KG coverage—tasks that the evidence-centered visual interface is designed to support.

Table 3: Near-miss and failure cases from the broad fault-set TEP evaluation. Candidate names are abbreviated; full identifiers use the `faultanchor:` prefix in artifact files.

| Fault | Expected                | Top-1                | Interpretation                                 |
|-------|-------------------------|----------------------|------------------------------------------------|
| 10    | stream_4.feed.temp.     | stream_2.feed.temp.  | Near miss; expected at rank 3, path retrieved. |
| 15    | sep_coolant.valve       | stream_2.feed.temp.  | Near miss; expected at rank 2, path retrieved. |
| 18    | condenser.heat.transfer | stream_2.feed.temp.  | Miss under current assets and features.        |
| 20    | Fault20                 | multi_valve.stiction | Asset-coverage gap; no Root-KGD anchor found.  |

## 6.2 Case 2: Wafer Defect Traceability

**Task and evidence boundary.** The wafer case evaluates whether RootLens can organize wafer spatial-pattern evidence into source-grounded, inspectable traceability paths. We use WM811K-style records [?] to test evidence normalization, entity linking, consistency checking, and path provenance. Public WM811K pattern labels are spatial-pattern annotations, not verified process root-cause references; this case therefore reports traceability coverage and reviewable candidate hypotheses, not process-RCA accuracy.

We separate two evidence layers with distinct purposes. The first uses classifier-produced evidence over five local Loc cases to evaluate whether predicted spatial-pattern evidence can be converted into KG-linkable observations and connected to candidate paths. The second uses native-label sampling across all eight supported public WM811K pattern classes to evaluate pattern coverage and traceability coverage; it is not a classifier-performance evaluation.

**Layer 1: classifier-evidence traceability over five Loc cases.** The classifier-evidence layer contains five local WM811K cases, all from the Loc pattern class. For records containing both native and predicted pattern metadata, native-vs-predicted agreement is 1.00. All five cases link to wafer evidence and return candidate paths, giving 1.00 traceability path coverage. The mean classifier confidence for Loc is 0.6613 and the mean number of linked KG entities is 4.0. The top candidate hypothesis is `ProcessNonuniformity`, supported by a source-grounded but low-confidence wafer-process interpretation. This result should be read as traceability over a narrow local sample, not as dataset-level WM811K performance. The classifier confidence is propagated as a source-quality cue, allowing analysts to weigh uncertainty alongside the candidate path rather than accepting the top hypothesis at face value.

**Layer 2: native-label pattern-stratified traceability.** The stratified layer samples one native-labeled row from each of the eight supported public WM811K pattern classes: `Center`, `Donut`, `Edge-Loc`, `Edge-Ring`, `Loc`, `Random`, `Scratch`, and `Near-full`. All eight patterns are covered, with mean linked entities of 4.0, mean consistency of 0.925, and traceability path coverage of 1.00. Exact pattern accuracy is not reported because this layer uses native-label sampling for coverage and source-grounded traceability, not classifier evaluation.

Table 4 summarizes the per-pattern results. Most patterns obtain consistency 1.0; `Random` and `Scratch` score 0.7, indicating evidence-to-KG relation conflicts. In the review workflow, an analyst can inspect the linked wafer-pattern observations, identify which relation checks failed, and examine the source edges supporting the candidate path—illustrating traceable candidate-path review rather than verified process RCA.

Two checked fixture cases (`wm811k.fixture.clean_nearfull` and `wm811k.fixture.descriptor_fallback`) confirm end-to-end reproducibility: both return consistency 1.0, four linked enti-



ties, five paths, and the top candidate `GlueRemovalInsufficient` sourced from `wafer_thesis`. These are candidate or plausible explanations only, not verified root-cause labels.

### 6.3 Case 3: Visual Evidence and KG Completion

**Task and evidence boundary.** The MVTec case evaluates visual evidence normalization and plausible KG-based explanation review. Detector outputs—including anomaly score, heatmap or mask geometry, morphology, location, severity, and confidence—are converted into evidence observations and linked to KG entities. MVTec-style data [?] provides inspection labels and localization signals, but not verified factory root-cause labels. Candidate paths are therefore framed as plausible explanation paths or KG-completion candidates, not ground truth.

**Transparent object selection.** We selected cable as the paper-facing visual evidence object using a criterion-weighted selection procedure over a calibrated local MVTec run [VERIFY: final detector and sampling protocol]. The selection combines three dimensions: visual model behavior (correct good/defect separation), explainable-path coverage, and KG entity coverage. As shown in Table 5, cable obtains the highest selection score, 0.7939, because it combines broad defect coverage (nine records, including eight defect records and one good record), correct good/defect separation (visual score 1.00), strong path coverage (0.8889), and adequate KG entity coverage (KG score 0.8333). Mean IoU is reported only as a localization signal for evidence extraction; it is not a detector-performance benchmark or an RCA metric. The candidate explanation targets for cable—including `AssemblyError`, `ContaminationCause`, `MechanicalContact`, and `MissingComponent`—are diverse and semantically reviewable. `Metal_nut` is retained as a secondary option with fewer records but high KG and path coverage.

**Visual evidence and plausible path review.** Two checked fixture cases (`mvtec_fixture_clean_scratch` and `mvtec_fixture_mask_fallback`) illustrate the full evidence-to-review path. Both obtain consistency 1.0, four linked entities, three candidate paths, and the top candidate `MechanicalContact` with source `manual_curation`. The analytical value of this case is not that `MechanicalContact` is a verified factory cause, but that localized visual evidence—heatmap-derived morphology, mask geometry, location descriptors—is normalized into reviewable observations, linked to KG entities, and traced to source-grounded paths. An analyst can inspect the `manual_curation` source edge to see what curated evidence supports the relation before accepting the candidate as a review target or flagging it as a KG-completion opportunity.

### 6.4 Expert Feedback

This subsection describes a planned formative evaluation protocol. Expert sessions have not yet been completed; participant details, procedure, and findings are [TO BE FILLED] and will be reported only after the study is conducted.

**Planned study.** We plan to recruit [VERIFY: number] participants with [VERIFY: roles and domain expertise] and ask them to inspect representative TEP, wafer, and MVTec cases using RootLens. The protocol compares analyst experience with isolated detector outputs against the full RootLens workflow, which adds unified evidence, source-edge provenance, candidate paths, consistency information, correction candidates, and KG review targets. Procedure details—tasks, duration, data shown, and collection method (interview, think-aloud, questionnaire, or mixed)—are [VERIFY].

**Evaluation questions and reporting plan.** The study is designed to assess four dimensions: whether the unified evidence schema helps compare heterogeneous anomaly outputs across image, process, and wafer cases; whether source-edge provenance makes candidate paths more inspectable as hypotheses; whether consistency checks and correction candidates expose evidence-to-KG mismatches; and whether append-only feedback is appropriate for controlled KG review in industrial RCA settings. We will not use expert feedback to claim causal correctness of RCA outputs. Feedback remains review state for paths, edges, entity links, correction candidates, and KG draft items, and does not automatically mutate production KG artifacts in the current system.

## 7 DISCUSSION AND CONCLUSION

### 7.1 Discussion

### 7.2 Conclusion

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Table 4: Pattern-stratified WM811K traceability layer. One native-labeled row per supported public pattern class. This layer evaluates pattern coverage and traceability path coverage, not classifier accuracy or process-RCA accuracy. Consistency below 1.0 indicates evidence-to-KG conflicts warranting analyst review.

| Pattern   | Cons. | Linked | Path cov. | Top candidate hypothesis |
|-----------|-------|--------|-----------|--------------------------|
| Center    | 1.0   | 4.0    | 1.0       | ProcessInterruption      |
| Donut     | 1.0   | 4.0    | 1.0       | ProcessNonuniformity     |
| Edge-Loc  | 1.0   | 4.0    | 1.0       | ChamberContamination     |
| Edge-Ring | 1.0   | 4.0    | 1.0       | EdgeProcessIssue         |
| Loc       | 1.0   | 4.0    | 1.0       | ProcessNonuniformity     |
| Random    | 0.7   | 4.0    | 1.0       | ParticleContamination    |
| Scratch   | 0.7   | 4.0    | 1.0       | HandlingScratch          |
| Near-full | 1.0   | 4.0    | 1.0       | GlueRemovalInsufficient  |
| Mean      | 0.925 | 4.0    | 1.00      | —                        |

Table 5: MVTec object-selection summary (top six objects shown). The selection score combines visual behavior, explainable-path coverage, and KG entity coverage; it is not an RCA accuracy metric. Mean IoU reflects localization signal for evidence extraction only. KG score measures entity coverage of the scenario-scoped knowledge graph.

| Object    | Rec. | Def. | Visual | IoU    | Path   | KG     | Score  |
|-----------|------|------|--------|--------|--------|--------|--------|
| cable     | 9    | 8    | 1.00   | 0.2748 | 0.8889 | 0.8333 | 0.7939 |
| metal_nut | 4    | 3    | 1.00   | 0.2819 | 0.7500 | 0.8750 | 0.7689 |
| carpet    | 6    | 5    | 1.00   | 0.3865 | 0.6667 | 0.8333 | 0.7606 |
| grid      | 6    | 5    | 1.00   | 0.1763 | 0.8333 | 0.8333 | 0.7603 |
| capsule   | 6    | 5    | 1.00   | 0.1967 | 0.6667 | 0.9167 | 0.7393 |
| zipper    | 8    | 7    | 1.00   | 0.3857 | 0.3750 | 0.9375 | 0.7084 |

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